

Day 19: Hugging Face Practical (Hands-on Implementation)

1. What is Hugging Face (Quick Recap)

Hugging Face provides:

- Pretrained models (GPT, BERT, T5, etc.)
- Transformers library
- Tokenizers
- Model hub (ready-to-use models)

 Goal: Use powerful LLMs without training from scratch

2. Installation & Setup

Install Required Libraries

Bash

```
pip install transformers  
pip install torch
```

 Optional:

Bash

```
pip install sentencepiece  
pip install accelerate
```

3. Pipeline API (Easiest Way to Use Models)

What is Pipeline?

- High-level API
- One line → full model execution

Example 1: Sentiment Analysis

Python

```
from transformers import pipeline

classifier = pipeline("sentiment-analysis")
result = classifier("I love Generative AI!")

print(result)
```

 Output:

Code

```
[{'label': 'POSITIVE', 'score': 0.99}]
```

Example 2: Text Generation (GPT)

Python

```
generator = pipeline("text-generation")
print(generator("AI is the future of", max_length=30))
```

Example 3: Summarization (T5/BART)

Python

```
summarizer = pipeline("summarization")

text = """Artificial Intelligence is transforming industries..."""
print(summarizer(text, max_length=50))
```

✅ Example 4: Question Answering

Python

```
qa = pipeline("question-answering")

context = "Hugging Face is a company that builds NLP tools."
question = "What does Hugging Face build?"

print(qa(question=question, context=context))
```

🧩 4. Under the Hood (Tokenizer + Model)

Pipeline is simple, but internally:

👉 Flow:

Code

Text → Tokenizer → Token IDs → Model → Output → Decode

✅ Manual Implementation (Important for Interviews)

Python

```
from transformers import AutoTokenizer, AutoModelForSequenceClassification
import torch

tokenizer = AutoTokenizer.from_pretrained("distilbert-base-uncased")
model = AutoModelForSequenceClassification.from_pretrained("distilbert-base-uncased")

inputs = tokenizer("I love AI", return_tensors="pt")
outputs = model(**inputs)

print(outputs.logits)
```

💡 Key Concepts

- **Tokenizer** → converts text → numbers
- **Model** → processes numbers
- **Logits** → raw output scores





5. Using Different Models (Real Mapping)

Task	Model Type	Example
Text Generation	GPT	Story writing
Classification	BERT	Sentiment
Summarization	T5/BART	News summary
Translation	T5	English → French



6. Hugging Face Model Hub



Huge collection of pretrained models

Example:

Python

```
pipeline("text-generation", model="gpt2")
```



You can:

- Search models
- Use directly
- Fine-tune later



7. Important Parameters (Very Important)

For Text Generation:

Python

```
generator("AI is", max_length=50, temperature=0.7, top_k=50)
```

- max_length → output size
- temperature → creativity
- top_k → randomness control



1. What is pipeline?

Python

```
from transformers import pipeline
```

👉 **pipeline = shortcut / high-level API**

💡 **Meaning:**

- Combines everything internally:

Plain text

Text → Tokenizer → Model → Output

👉 You don't need to manage anything manually

✅ **Example:**

Python

```
pipeline("sentiment-analysis")
```

👉 Automatically loads:

- tokenizer
- model
- config

📁 **What to say:**

"Pipeline is the easiest way to use pretrained models in just one line."

2. What is AutoTokenizer?

Python

```
from transformers import AutoTokenizer
```

👉 **Tokenizer = converts text → numbers**

💡 **Why needed?**

Models don't understand text ❌

They understand **numbers (tokens)** ✅

🔄 **Example:**

Plain text

```
"I love AI"  
→ ["I", "love", "AI"]  
→ [101, 2345, 6789]
```

✅ **Code:**

Python

```
tokenizer = AutoTokenizer.from_pretrained("google/flan-t5-small")
```

👉 "Auto" means:

- Automatically selects correct tokenizer for model

📁 **What to say:**

"Tokenizer converts human language into a format that the model can understand."

3. What is AutoModelForSeq2SeqLM?

Python

```
from transformers import AutoModelForSeq2SeqLM
```

👉 This is the **actual AI model**

💡 **Meaning:**

- Seq2Seq = **Input → Output transformation**
- Used for:
 - Summarization
 - Translation
 - Q&A

✅ **Code:**

Python

```
model = AutoModelForSeq2SeqLM.from_pretrained("google/flan-t5-small")
```

👉 Loads pretrained T5 model

🎬 **What to say:**

"This is the brain of the system that processes input and generates output."

4. What is `from_pretrained()`?

Python

```
.from_pretrained("google/flan-t5-small")
```

👉 Loads model from Hugging Face hub

👉 No training needed ❌

👉 Ready-to-use model ✅

🎬 What to say:

"We are using a pretrained model, so we don't need to train from scratch."

5. What is `return_tensors="pt"`?

Python

```
inputs = tokenizer(text, return_tensors="pt")
```

👉 Converts output into **PyTorch tensors**

💡 Why?

Model expects:

Plain text

numbers → tensor format

🎬 What to say:

"We convert tokens into tensors so the model can process them."



6. What is `model.generate()`?

Python

```
outputs = model.generate(**inputs)
```

👉 This is where **actual prediction happens**

💡 **Meaning:**

- Model reads input
- Predicts next tokens
- Generates output

🎬 **What to say:**

"The model generates output based on learned patterns."

7. What is `tokenizer.decode()`?

Python

```
tokenizer.decode(outputs[0], skip_special_tokens=True)
```

👉 Converts numbers → human-readable text

🔄 **Flow:**

Plain text

Output IDs → Text

🎬 **What to say:**

"We convert model output back into readable text."

